No. of Functional Features Included in the Solution

Date	30 June, 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Functional Features Overview

This document lists the functional features implemented in the Smart Agricultural Production Optimization Engine (OptiCrop) project. The system leverages Machine Learning and a Flask-based web application to provide intelligent crop recommendations and environmental suitability analysis.

The application processes agricultural parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to assist farmers, researchers, and policymakers in making data-driven decisions.

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status (Done / in progress / Pending)	Marks Contribution
1	Soil & Environmental Data Input	Accepts input values such as N, P, K, temperature, humidity, pH, and rainfall through a web form.	Flask UI	Done	High
2	Data Preprocessing	Cleans and prepares input data for prediction	Data processing	Done	High
3	Crop Recommendation System	Predicts the most suitable crop based on given environmental conditions.	Prediction Engine	Done	High
4	Model Integration	Loads trained machine learning model using Pickle.	Backend	Done	Medium
5	User-Friendly Web Interface	Displays prediction results through an interactive web page..	Frontend	Done	Medium
6	Dataset Handling & Model Storage	Manages dataset and serialized ML model file.	Storage	Done	Low

Feature Summary:

Metric	Count / value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	Data Input, Data Preprocessing, Crop Recommendation, Model Integration
Additional / Nice-to-have Features	User Interface, Error Handling, Environmental Analysis
Features Tested & Verified	All features tested successfully

Feature Category Breakdown

S.No	Category	Features in Category	Example Feature
1	User Interface(UI)	2	Input Form, Result Display Page
2	Backend / Logic	3	Data Preprocessing, Prediction Engine, Suitability Analysis
3	Database / Storage	1	Dataset Management and Model Storage
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	1	Input Validation and Exception Handling